

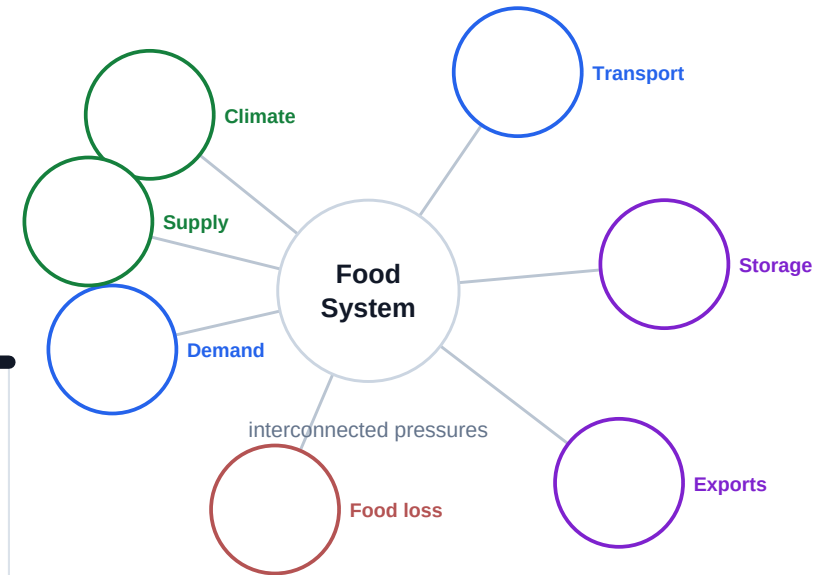
# Systems Under Pressure

## The Mathematics of Food Systems

**"All models are wrong, but some are useful."**

George Box

A concise investigation brief for studying real food systems under mathematical pressure.



# A model is useful, but incomplete.

You are not simply completing Algebra 2 tasks. You are investigating an unstable food system and using mathematics to make its behavior more visible. Models reveal patterns, hide variables, depend on assumptions, and eventually break down. Strong mathematical thinkers name those limits honestly.

## Models reveal

patterns, rates, structure, constraints, thresholds, and possible futures

## Models hide

unmeasured variables, human decisions, sudden disruptions, uncertainty, and local context

## Models change

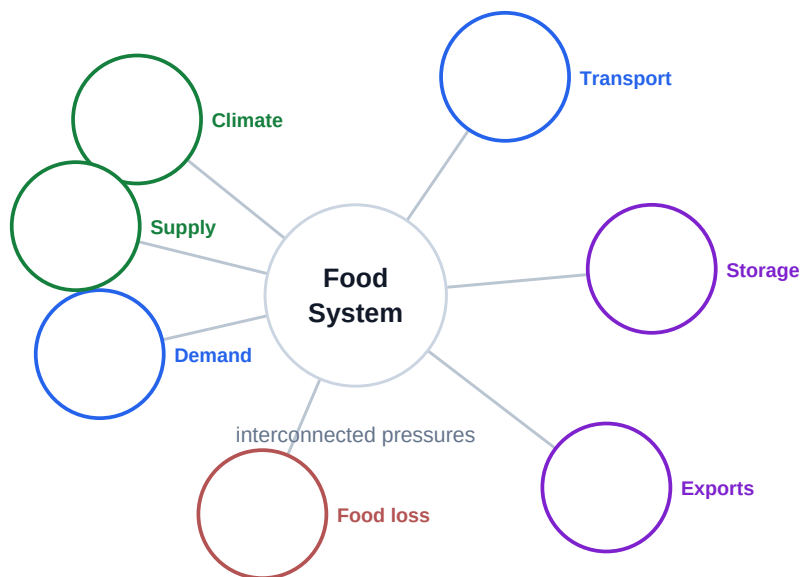
when assumptions, rates, dimensions, domains, or constraints are revised

## Revision is evidence of thinking

Crossed-out assumptions, recalculated thresholds, revised graphs, changed interpretations, and honest limitations are not mistakes to hide. They are evidence that the investigation became more sophisticated.

# One system. Many pressures.

Food systems are interconnected. Changes in one part can reshape the whole system.



## Mathematics helps us

Model change, identify patterns, compare possibilities, estimate outcomes, critique pre

## Strong thinkers ask

What assumptions are we making? What variables matter most? Where does the mod

# Choose the format that best communicates the investigation.

The purpose is communicating mathematical reasoning, not producing a polished slideshow.

## Allowed formats

Slides, filmed discussions, workbook walkthroughs, digital whiteboards, screencasts, annotated journals, documentary-style explanations, or hybrid combinations.

## Math remains central

Models, graph analysis, comparisons, assumptions, revisions, limitations, final claims, and visible mathematical evidence.

## Clarity over polish

Do not prioritize cinematic editing, effects, overproduction, or memorized performance. Prioritize reasoning and explanation.

## Workbook walkthroughs are encouraged

The journal is a major part of the investigation. Students may explain crossed-out assumptions, revised graphs, recalculations, and changing interpretations directly from the workbook.

# What makes the investigation visible

These pieces work together. The journal shows the evolving inquiry; the final communication explains the mathematical story.

## Investigation Journal

Rough work, sketches, Desmos screenshots, calculations, AI interactions, revisions, and evolving reasoning.

## Mathematical Artifacts

Graph annotations, recursive calculations, polynomial long division, tables, recalculations, and revised models.

## Group Reflection Vlogs

Short mathematical conversations showing uncertainty, revision, disagreement, critique, and progress.

## Final Communication

The chosen format that communicates models, comparisons, assumptions, limitations, and claims.

# One shared investigation. Visible individual thinking.

Do not divide units by student. The group functions as a collaborative mathematical inquiry team.

## Collaborative work

Students think together, discuss together, graph together, revise together, compare together, critique together, and synthesize together.

## Individual evidence snapshots

Each student contributes smaller, frequent artifacts: graph interpretations, model comparisons, recalculations, AI critique, revised assumptions, or limitation reflections.

## Every student speaks mathematics

Every student must visibly contribute reasoning, interpretation, comparison, critique, or explanation in group vlogs and final communication.

# Choose one documentary investigation prompt.

Each scenario begins with a human tension. Your group narrows it into one measurable mathematical investigation.

## A | United States

The \$5 Footlong Isn't \$5 Anymore

Changing everyday food costs reshape affordability.

**Key tensions:** inflation, pricing, shrinkflation, labor, transportation

## B | Global

When War Changes the Price of Bread

Disruption in one region can shift prices and recovery patterns elsewhere.

**Key tensions:** exports, wheat, transport, recovery, supply chains

## C | Sub-Saharan Africa

What If Food Exists - But Never Arrives?

Food can be produced and still be lost before reaching people.

**Key tensions:** spoilage, refrigeration, storage, infrastructure

## D | Colombia

Can Climate Change the Future of Coffee?

Environmental and economic changes can alter yield, exports, and thresholds.

**Key tensions:** rainfall, coffee yield, climate, transport, exports

# A good question is narrow enough to model.

Avoid giant topics. Look for one changing quantity, one measurable relationship, and one meaningful constraint.

## Too broad

How inflation affects food  
Climate and coffee  
Food insecurity  
Food waste

## Better

How sandwich prices changed over time  
Rainfall vs coffee yield in one region  
Food spoilage during transport stages  
Food remaining after repeated losses

## Strong focus test

Can your group identify one main changing quantity, one measurable relationship, one prediction or threshold question, one physical or cost constraint, and one connection across mathematical lenses?



# The five-class arc is one evolving investigation.

Each phase builds on the same system. The mathematics becomes more sophisticated through comparison and revision.

## 1. Entering the System

Notice relationships before choosing equations. Move from topic choice to a measurable system relationship.

## 2. Modeling Change

Represent repeated change with terms, formulas, and accumulated totals. Predict and test reliability.

## 3. Thresholds

Model compounding, decay, and threshold moments. Interpret model breakdown.

## 4. Constraints

Use structure and graph behavior to understand capacity, tradeoffs, extrema, and restricted domains.

## 5. Synthesis

Turn separate models into one defensible mathematical claim about uncertainty.

# Different structures reveal different things.

Students repeatedly answer: Why this model? What does it reveal? What does it miss? How do I know?

## Sequences & Series

How is the system changing over time?  
Compare discrete vs continuous,  
arithmetic vs geometric, recursive vs  
explicit, finite vs infinite.

## Exponentials & Logs

When does change accelerate, decay,  
compound, or cross a threshold?  
Compare linear vs exponential and  
sampled values vs continuous models.

## Polynomials

How do capacity, cost, and physical  
constraints reshape the system? Analyze  
structure, end behavior, extrema, and  
domain.

## Lightweight model fit

Use  $r$  and  $R^2$  as credibility and critique tools. Interpret direction, strength, fit, hidden variables, and causation limits. This is not a full statistics unit.

# Three content standards. Three skill grades.

The final rubric separates Algebra 2 content from cross-cutting mathematical habits. Both matter.

## CORE ALGEBRA 2 CONTENT STANDARDS

### 1. Sequences and Series

See the rubric for performance descriptors. This standard focuses on the mathematical content of the investigation.

### 2. Exponentials and Logarithms

See the rubric for performance descriptors. This standard focuses on the mathematical content of the investigation.

### 3. Polynomials

See the rubric for performance descriptors. This standard focuses on the mathematical content of the investigation.

## CROSS-CUTTING SKILL GRADES

### 4. Mathematical Communication and Interpretation

Assessed across the whole investigation as a mathematical habit: communication, comparison, critique, revision, precision, and flexible thinking.

### 5. Comparing Models, Assumptions, and Limitations

Assessed across the whole investigation as a mathematical habit: communication, comparison, critique, revision, precision, and flexible thinking.

### 6. Algebraic Precision and Flexible Number Sense

Assessed across the whole investigation as a mathematical habit: communication, comparison, critique, revision, precision, and flexible thinking.

# SAIL L5: Co-Create

AI is a thinking partner, not an answer machine. It should deepen reasoning and make critique sharper.

## AI may help you

Challenge assumptions, critique models, identify missing variables, compare interpretations, explain  $r$  and  $R^2$ , organize ideas, and revise explanations.

## AI may not replace you

Do not submit reasoning you do not understand, ask AI to complete the project, hide meaningful AI use, or replace your own mathematical voice.

## Transparency sentence

AI use: We asked ChatGPT to critique our constant-growth assumption. We rejected one suggestion, revised our rate assumption, and recalculated the threshold.